

Getting the Most out of Virtual Reality: Evaluating Short Breaks to Reduce Cybersickness and Cognitive Aftereffects

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Figure 1: In-game screenshot sourced from Space Pirate Trainer Media Kit [17] and is in the public domain.

ABSTRACT

Considering the importance of getting the most out of virtual reality (VR) training, we evaluated the use of breaks to reduce virtual reality aftereffects in head-mounted displays. In a crossover design, we compared a 50-minute no-break exposure with an interrupted 50-minute exposure with four 10-minute breaks. Cybersickness, reaction times, workload and spatial learning were measured. Despite device manufacturers recommending breaks to reduce sickness, we found no evidence that breaks can mitigate aftereffects. Introducing breaks may also impact performance and logistical costs for training.

Keywords: Simulator sickness, Motion sickness, Simulation and training, Virtual environments

Index Terms: Human-centered computing—Human computer interaction—Interaction paradigms—Virtual reality

1 INTRODUCTION

Virtual reality (VR) technologies are disrupting industries worldwide. With greater accessibility of head-mounted displays (HMDs) and advancements in hardware, many organisations are increasingly using VR for training. However, there are still several outstanding questions on how best to structure and implement VR training. Crucially, questions around the length of VR exposure are often at the core of implementing training procedures.

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Negative symptoms such as VR sickness may hinder the potential benefits of VR training. A recent systematic review and meta-analysis on self-report VR sickness estimates that around 15.6% of people will drop out of VR studies due to sickness. On average, many people who complete these studies will still experience a high level of VR sickness [28]. The length of each VR exposure will also have an impact on VR sickness [5, 28, 33]. Hence, developers often make design choices to limit VR sickness in users, such as innovative locomotion techniques [18], rest frames [1, 35], visual blurring [23], and viewpoint control [8]. Implementing these design choices may depend on the characteristics of the simulation and skill set of the VR development team. Strategies employed by the user can also help to mitigate VR sickness during training [7, 16, 19, 30]. The most straightforward user-directed strategy is to introduce repeated breaks during training exposures.

Device manufacturers of HMDs recommend taking short breaks in their safety guidelines to mitigate VR sickness symptoms [7, 16, 30]. Most manufacturers suggest breaks of 10-15 minutes between VR exposures (ranging from 30 mins to 1 hour), while some do not specify, leaving it up to the user. The device safety recommendations rely on the user's ability to self-assess symptoms of VR sickness. Some side-effects such as changes in vision and cognition [2, 14, 34] may be difficult for users to notice. However, it is unclear what impact changes in vision and cognition may have on subsequent behaviours outside of VR. There is limited publicly available research demonstrating the impact of breaks on repeated VR exposure and whether it is an effective strategy for mitigating aftereffects.

There are no studies that directly compare cognitive aftereffects and cybersickness between long continuous VR exposures to exposures with breaks. Notably, some studies report that scheduled

breaks in between an intervention will increase self-reported symptoms with each subsequent exposure [4,22]. These reports are consistent with the research showing that self-reported symptoms increase with longer uninterrupted exposures [5,24,28]. When a person puts on a VR headset, the real world is blocked out, and their oculomotor system needs to adapt to the new visual and spatial demands of the virtual environment in the headset. This process of adaptation also leads to changes in visual perception, hand-eye coordination, and balance [2,26,34]. Both adaptation and re-adaptation are related to VR aftereffects [2,31]. Accordingly, repeated exposures within a short time frame involves adaptation and re-adaptation to the real and simulated environment, which may lead to an accumulation of aftereffects.

There is some evidence to suggest that habituation may be beneficial for reducing aftereffects in HMDs, but these studies typically have intervals of at least 24 hours between exposures [15]. Having short breaks during VR training could be beneficial for learning, as it has been demonstrated to help cognitive performance in other contexts [32]. Given the importance of getting the best out of VR training, this study investigated the impact of breaks on VR sickness, perceived workload, reaction time, and learning. These aims were addressed with a within-subjects design comparing aftereffects between a 50-minute continuous exposure (i.e. no breaks) with an interrupted 50-minute exposure comprising of four 10 mins breaks. To determine which exposure condition resulted in negative effects, VR sickness and changes in cognition were monitored throughout the experiment.

2 METHODS

2.1 Participants

Twenty-five participants ($m = 16; f = 9$) aged 18 to 35 years took part in the study ($M = 23.13, SD = 4.50$). All participants had normal to corrected-to-normal visual acuity and a stereoacuity of at least 63 arcseconds measured via the Butterfly Stereo Acuity test. If participants required prescription glasses, they were advised to use them during the experiment. Participants were given an honorarium of \$20/hour for their time. This research complied with the Declaration of Helsinki and was approved by the University of South Australia's Human Research Ethics Committee. Informed consent was obtained from each participant.

2.2 Materials and Apparatus

2.2.1 Virtual Reality Setup

An HTC Vive Pro HMD (HTC Corporation) presented a popular VR experience titled Space Pirate Trainer (developed by I-Illusions, Belgium). The HMD was tethered to a high-performance computer with an Intel Quad Core i7-7820HK processor at 2.90GHz, 16 GB RAM and a Nvidia Geforce GTX 1080 8GB graphics. Space Pirate Trainer required users to shoot waves of flying targets using different weapons. Users could dodge projectiles by moving in three dimensions and were encouraged to achieve a high score Fig. 1.

2.2.2 Simulator Sickness Questionnaire (SSQ)

The SSQ [21] is the most widely used self-report measure of simulator sickness [27,28], and was employed in this study to measure VR sickness. The SSQ consists of 16 items with a 4-point scale measuring symptom severity, ranging from 0 (none) to 3 (severe). The SSQ has three subscales, nausea, oculomotor disturbance and disorientation. Accordingly, each subscale and total symptom ratings are weighted as advised by [21].

2.2.3 NASA Task Load Index (NASA-TLX)

The NASA-TLX is a self-report questionnaire developed to measure the subjective workload of a given task [13], and more recently, it has been used to detect task load differences from using HMDs [6]. The TLX contains six subscales: mental, physical and temporal

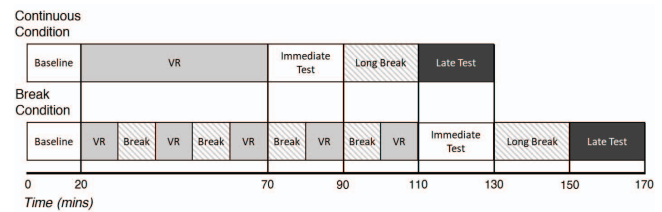


Figure 2: Showing the study design for both continuous and break conditions completed on different days. Each day consisted of a baseline, immediate test, break and late test. In the break condition, participants completed five 10-minute VR exposures and four 10-minute breaks.

demand; and overall performance, effort and frustration levels. In our study, participants used the electronic version and moved a red line presented on an iPad touchscreen. This line intersects the 100-point scale, indicating their subscale level from 0 to 100. Weighted scores are gathered by multiplying the number of times a subscale was rated above another (maximum of 5) by the raw individual subscale rating out of 100. Total scores are calculated using the mean of all weighted scores as stipulated in [10] and referred to as weighted ratings.

2.2.4 Cognitive Tasks

Two Cambridge Neuropsychological Test Automated Battery (CANTAB) cognitive tasks (Cambridge Cognition), namely the 5-choice reaction time task (RTI) and the paired associates learning task (PAL), were administered on an iPad 2 using the CANTAB app. The RTI requires participants to monitor 5 locations on the screen and respond as quickly as possible when a target appears. To do this, a participant must press and hold a button at the lower half of the screen while waiting for a target (i.e. yellow dot) to appear in 1 of 5 locations above. When the target appears, the participants must release the button and touch the target stimulus as quickly as possible. The outcome measures for the RTI are decision and movements speeds. Only correct responses are included in these calculations. Decision speed is the median duration from the time the target stimulus appears and the participant releases the button. The movement speed is the median duration of the release of the button to touch the target stimulus. In the PAL task, multiple boxes are displayed in a circular pattern and are "opened" in a randomised order. Some boxes will display a unique pattern when opened, and participants must wait and memorise all given patterns before identifying the location of each pattern. When it is the participants turn to respond, a pattern is displayed in the centre of the screen and participants must identify the location they saw that pattern by touching the box it appeared in. Throughout the task, the difficulty is increased by including additional boxes and patterns to identify. The outcome measures include the first attempt memory score and total errors. The PAL took approximately 8 mins to complete.

2.3 Procedure

A crossover design with two VR exposure conditions was conducted across two separate days Fig. 2. On one day, participants completed an uninterrupted continuous 50-minute session of VR, and on the other day, they completed a 50-minute session with four breaks. The order of sessions was counterbalanced across participants. Upon arrival at the VR laboratory, participants were guided through the consent process and given verbal instructions on what to do. Once consented, all participants underwent stereoacuity screening using the Butterfly Stereo Acuity test (Vision Assessment Corporation, 2007). A conservative cut-off of 63 arcseconds was used to reduce the likelihood of visual disturbances from poor stereopsis. Eligible participants were allocated to either start with the uninterrupted or

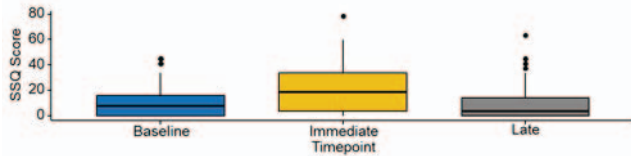


Figure 3: Boxplots of SSQ showing the differences between timepoints before VR exposure (baseline), immediately after 50 mins of VR, and 40 mins after VR (late).

interrupted session. There was a minimum of 24 hours between sessions to reduce the influence of residual aftereffects from the previous exposure. Each session consisted of a baseline, 50-minute VR exposure, immediate test, break and late test. An Apple iPad was used to administer the SSQ, NASA-TLX, and CANTAB RTI and PAL tasks. During the baseline, participants completed visual screening, RTI task, SSQ and static balance assessment. Participants then completed either a continuous or break VR exposure condition. For the continuous exposure, participants engaged with VR for 50 minutes without any interruptions. In the break condition, participants completed five 10-minute VR exposures with four 10-minute breaks in between (see Fig. 2). During the breaks, participants completed static balance measures which are not assessed in this paper but reported in [3], and the SSQ. After completing a session, participants were then administered immediate tests, including the SSQ, NASA-TLX, RTI and balance measures. After these immediate tests, participants were given a 20-minute rest period, and late tests were completed 40-minutes after exiting VR. Late tests included SSQ, balance, RTI, and PAL. Total experimental times combined for both sessions was about 5 hours.

3 RESULTS

3.1 SSQ

A mixed model repeated measures ANOVA was used to compare timepoint (baseline, immediate, and late) as within-subject factors and exposure condition (continuous, and break) as between-subject factors.

For total SSQ scores, there was a significant main effect of timepoint [$F(2,46) = 13.932, p < .001, \eta_p^2 = .377$] with higher sickness scores were reported immediately after VR ($M = 20.03, SE = 3.04$) compared with the baseline ($M = 10.75, SE = 3.04$) [$t(23) = 4.339, p < 0.001, d = .886$] and late measures ($M = 10.28, SE = 3.04$) [$t(23) = 4.207, p < .001, d = .859$] (see Fig. 3). Exposure conditions [$F(1,23) = 0.236, p = .632, \eta_p^2 = .010$] had no effect, and there were no significant interaction effects either [$F(2,46) = 1.208, p = .308, \eta_p^2 = .050$].

For nausea subscale scores, there was a significant effect of timepoint [$F(2,46) = 9.916, p < .001, \eta_p^2 = .301$] with greater nausea scores immediately after VR ($M = 20.074, SE = 3.119$) compared with the baseline ($M = 10.335, SE = 3.119$) [$t(23) = 3.829, p < .001, d = .782$] and late measures ($M = 10.136, SE = 3.119$) [$t(23) = 3.65, p < .001, d = .745$]. There was no effect of exposure [$F(1,23) = 1.243, p = .276, \eta_p^2 = .005$], and the interaction was also not significant [$F(2,46) = .493, p = .614, \eta_p^2 = .021$].

For oculomotor subscale scores, there was a significant effect of timepoint [$F(2,46) = 13.024, p < .001, \eta_p^2 = .362$] with higher oculomotor scores immediately after VR ($M = 24.003, SE = 3.950$) compared with the baseline ($M = 12.633, SE = 3.950$) [$t(23) = 2.453, p = .022, d = .501$], but were not different to late measures ($M = 14.212, SE = 3.950$) [$t(23) = 1.04, p = .311, d = 0.212$]. There was no effect of exposure [$F(1,23) = .024, p = .877, \eta_p^2 =$

.001] and the interaction was not significant either [$F(2,46) = 1.618, p = .209, \eta_p^2 = .066$]. For disorientation subscale scores, there was a significant effect of timepoint [$F(2,46) = 11.578, p < .001, \eta_p^2 = .335$] with greater disorientation scores in immediately after VR ($M = 29.290, SE = 5.048$) compared with the baseline ($M = 17.400, SE = 5.048$) [$t(23) = 3.357, p = .003, d = .685$] and late measures ($M = 11.600, SE = 5.048$) [$t(23) = 3.75, p = .001, d = .765$]. There was no effect of exposure [$F(1,23) = .129, p = .722, \eta_p^2 = .006$] and the interaction was not significant either [$F(2,46) = 1.095, p = .343, \eta_p^2 = .045$].

3.2 NASA-TLX

Weighted NASA-TLX scores for each of the six indices and overall rating were used in a paired samples t-tests comparing the continuous and break VR conditions. Wilcoxon W (Z) is reported for comparisons that violate Shapiro-Wilk (W) normality tests (see Table 1). Only the task load scale for performance was significant. Participants reported that the break condition had a greater negative impact on their perceived performance in the game than the continuous exposure.

3.3 Cognitive Tasks

One participant was excluded from the analyses of the cognitive measures due to technical failure resulting in missing data. The CANTAB 5-choice RTI separates decision times and movement speeds comprising of reaction times. These components of reaction time were analysed separately in a mixed model repeated measures ANOVA to determine which component may be affected by exposure conditions. For decision times, there was no significant effect of exposure [$F(1,22) = .121, p = .731, \eta_p^2 = .005$], time point [$F(2,44) = .733, p = .486, \eta_p^2 = .032$] and the interaction [$F(2,44) = .108, p = .898, \eta_p^2 = .005$] was not significant either. For movement speeds, there was no effect of exposure [$F(1,22) = 1.560, p = .225, \eta_p^2 = .066$], time point [$F(2,44) = .519, p = .599, \eta_p^2 = .023$], and the interaction was also not significant [$F(2,44) = 1.186, p = .315, \eta_p^2 = .051$].

The CANTAB paired associates learning task (PAL) assesses spatial memory and new learning via two outcome measures comprising first attempt memory score and total errors. Notably, the PAL was only given to participants in the late timepoint to compare continuous and break conditions. The first attempt memory score captures the number of times a participant chose the correct box on their first attempt when recalling the pattern locations. A paired t-test compared continuous ($M = 17.43, SE = .457$) and break ($M = 17.30, SE = .63$) conditions to examine the impact on spatial memory and found no significant effect [$t(22) = .227, p = .822, d = .047$]. Examining the total errors of participants also led to no significant results [$t(22) = -.205, p = .839, d = -.043$].

4 DISCUSSION

This study aimed to investigate the impact of short breaks within a VR session on sickness and cognition after exiting VR. Despite device manufacturers recommending breaks to reduce VR sickness, we found no evidence to support this strategy in our study. If anything, repeated breaks impacted participants' perception of their in-game performance and would lead to higher logistical costs for training. These results question the usefulness of breaks in facilitating VR training as they did not show any obvious benefit in this study.

4.1 Do breaks mitigate VR sickness?

Overall, participants in this study demonstrated a high level of self-reported VR sickness immediately after VR in both conditions. This result is consistent with studies showing that longer exposures lead to a higher reporting of symptoms [5, 28, 33]. Self-reported sickness

Table 1: Comparisons between the continuous and break conditions on the NASA-TLX. Significance at $p < .05$.

Task Load Measure	Normality (Shapiro-Wilk)	Means (SE) of exposure conditions		Paired Comparison	Effect Size
		Continuous	Breaks		
Overall Rating	$W = .917, p = .050$	61.6 (2.8)	57.7 (3.1)	$t(23) = 1.370, p = .184$	$d = .280$
Mental	$W = .966, p = .576$	244.2 (23.7)	267.5 (18.8)	$t(23) = -.752, p = .460$	$d = -.154$
Physical	$W = .874, p = .006^*$	132.1 (29.8)	98.1 (26.4)	$Z(23) = 152.000, p = .417$	$r = .202$
Temporal	$W = .954, p = .327$	160.0 (26.2)	119.4 (27.1)	$t(23) = 1.704, p = .102$	$d = .348$
Performance	$W = .869, p = .005^*$	97.3 (16.7)	117.5 (16.0)	$Z(23) = 73.000, p = .049^*$	$r = -.471$
Effort	$W = .949, p = .261$	224.4 (28.2)	194.6 (23.8)	$t(23) = 1.101, p = .282$	$d = .225$
Frustration	$W = .861, p = .003^*$	67.1 (16.5)	68.3 (23.1)	$Z(23) = 118.000, p = .627$	$r = .129$

could not be explained by postural sway disturbances [3]. Symptom levels on the SSQ were categorised accordingly [20, 21], whereby a total SSQ score of 20 is considered serious symptoms of simulator sickness. Although participants in our study demonstrated VR sickness, the continuous and break conditions were not significantly different from one another. This finding suggests that 10-minute breaks had little impact on reducing symptoms of VR sickness.

Late measures were taken 40 minutes after exiting VR, whereby participants symptoms significantly decreased and returned to baseline levels. There were no differences in exposure conditions for the late measures. Although 10-minute breaks did not have a significant decrease on symptoms of VR sickness, the fact that participants self-report measures returned to baseline suggests that longer 40-minute breaks may be more beneficial for VR sickness. Determining an appropriate duration of breaks from VR is especially important for VR training. The findings from this study provide useful insights for industries and researchers employing training simulations in HMDs.

4.2 What impact do breaks have on perceived workload?

Participants did not perceive either of the exposure conditions to require more effort nor place higher demands on physical, mental or temporal dimensions. However, repeated breaks in the break condition were rated as having a negative impact on perceived performance. Perceived workload is an important aspect of training, and in many instances, breaks have been thought to improve cognitive performance [32]. Introducing repeated short breaks would have increased the total time of the break condition by 40 mins compared to the continuous condition. Any positive benefits of breaks may have been negated by the repeated adaptation between real and simulated environments thus impacting on the effectiveness of short breaks.

4.3 What impact do breaks have on aspects of cognitive performance?

Cognitive performance such as reaction times, mental load and learning outcomes are essential for many successful VR training simulations [9, 11, 12]. Various studies have reported changes in reaction times immediately after VR, but these findings are inconsistent. Some studies have demonstrated a negative effect (slower) on reaction times [25, 34], other studies have shown a positive effect (faster) on reaction times [29, 33]. Changes in reaction times are typically measured during or immediately after VR but not after a break. In our study, there was no impact of VR exposure conditions on the decision or movement speeds. Measurements comparing the baseline to late measurements were also not significant. These results suggest that there are no lasting effects of VR on reaction times.

4.4 Can breaks affect subsequent learning?

Using VR as a training tool offers many advantages; however, aftereffects from VR, such as cybersickness, sopite symptoms and fatigue, could continue to affect the user after training and negatively impact subsequent learning activities. To understand these effects on subsequent learning, we assessed spatial memory in a

paired associates learning task at the late time point. Examining learning at the late time point enabled us to observe the impact of pre-existing cybersickness on learning outcomes between the exposure conditions. There were no concerning effects on spatial learning between the continuous and break conditions. Spatial learning is one aspect of learning that could be impacted by cybersickness, and other types of learning outside the scope of this study, such as perceptual-motor learning or procedural learning, could also be impacted by aftereffects. Hence, it is essential that both VR training and aftereffect researchers consider the impact of cybersickness on subsequent learning.

4.5 Implications for training

This study found no observable differences between the continuous and break conditions. Notably, the breaks condition was 40 minutes longer than the continuous exposure, which would add to the logistics and costs of running VR training. For example, if an organisation wanted to train 50 people with four 10-minute breaks (50x40min) scheduled into their training, their overall training would be 33 hours longer than not having breaks. Hence, future research investigating the optimal scheduling for training has important practical implications. Exploring alternative user-directed strategies to mitigate sickness could lead to better outcomes and have greater practical advantages.

5 CONCLUSION

Device manufacturers of HMDs recommend taking short breaks in their safety guidelines to mitigate VR sickness symptoms, however, there is limited evidence to support this recommendation. To date, this is the first study to directly compare the impact of breaks on VR sickness, perceived workload, reaction time, and learning. Our findings do not support the hypothesis that short breaks can mitigate aftereffects. Longer rest periods of 40 minutes are sufficient to help reduce aftereffects from lengthy continuous exposures without breaks as well as aftereffects accrued from repeated exposure. Participants reported that repeated breaks impacted their perceived performance during VR compared to longer continuous exposures. More research is needed to investigate different break lengths in reducing aftereffects and optimising training schedules. Caution needs to be taken into consideration when touting breaks as a strategy to mitigate aftereffects because this approach may not work for everyone and may even have drawbacks on performance and add to the logistical costs of training. These learnings can be applied to the area of VR training, whereby organisations and users that are keen to implement training schedules may want to weigh up the practical impact of breaks on aftereffects and performance.

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